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Xuemin (Sherman) Shen *Editors*

5G Mobile Communications

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Preface

Mobile communications have been instrumental in transforming our contemporary societies in the past decades. From the first-generation (1G) of analogue mobile phone system to the newest commercial fourth-generation (4G) long-term evolution (LTE) networks deployed widely across the global, mobile communications have fundamentally changed the ways as to how humans in the modern society access, exchange, and share information with each other. Currently, we are at an era of mobile Internet with explosive big data. The growing demand for mobile data traffic and the proliferation of applications requiring high data rates have raised a significant interest in the definition of new standards in the mobile market. This calls for next-generation mobile communication systems, which should have to respond to an avalanche of traffic, an explosion in the number of connected devices, and the large diversity of use cases and requirements.

Against the above backdrop, the fifth-generation (5G) mobile communications are fast emerging to tackle the challenges brought by an exponential increase in wireless data traffic. On top of the massive increase in data volumes and rates, a formidable challenge for the 5G networks to deal with is how to connect billions of smart devices such as surveillance cameras, smart-home/grid devices, connected sensors, etc. The primary goals of 5G networks are to support a 1000-fold gain in capacity, connections for at least 100 billion devices, and 10 Gb/s delivered to individual users. Furthermore, new 5G networks will be able to provide mass low-latency and ultrareliable connectivity between people, machines, and devices, which will ultimately usher in the era of the Internet of Things (IoT). To meet these enormous challenges, disruptive innovations and drastic improvements need to be made in the mobile network architecture design in both the physical and upper layers.

The International Telecommunications Union (ITU) has stipulated 2020 to be the target year of standardising future 5G mobile networks. Although the detailed technical approaches to implementing 5G mobile networks remain uncertain at the time of this writing, several breakthrough 5G techniques stand out such as massive MIMO and millimetre-wave (mmWave) communications. This book aims to be one of the first comprehensive books to reveal the enabling techniques underpinning

next-generation 5G networks and to address the challenges and opportunities brought by 5G mobile communications. Specifically, the book is divided into three major parts: Part I Overview of 5G Networks, Part II Transmission and Design Techniques for 5G Networks, and Part III Networking Techniques and Applications for 5G Networks.

Part I of this book provides a comprehensive introduction to and overview of 5G networks. It consists of three chapters.

The chapter “An Overview of 5G Requirements” presents an overview of next-generation 5G mobile networks. To facilitate the study of 5G requirements and to provide guidance to 5G technical design, this chapter discusses several typical deployment scenarios including indoor hotspot, dense urban, urban macro, rural, and high-speed scenarios. It also presents high-level key capabilities and detailed technical requirements for 5G networks. Some technical performance metrics of 5G networks are also discussed.

The chapter “Spectrum Analysis and Regulations for 5G” discusses various aspects of 5G spectrum issues. It is expected that 5G requires much more bandwidth as well as more flexibility in spectrum usage and management. The suitable frequency ranges of 5G will include those bands below 6 GHz such as re-farmed 2G/3G spectrum, identified frequency bands for IMT, and also WRC-15 candidate bands. However, due to the scarcity of spectrum below 6 GHz, it is imperative to seek potential frequency ranges above 6 GHz. Controlled spectrum sharing is an important way of reusing spectrum to complement current licensed dedicated spectrum, which is still the foundation for the operation of 5G systems.

The chapter “Spectrum Sharing for 5G” first introduces spectrum sharing for 5G systems, which consists of multiple spectrum types with different scenarios. Then, spectrum sharing techniques mapped into different scenarios are introduced, i.e. coordination protocol, GLDB support, cognitive/DSA, and MAC-based coexistence. Besides, current applications of these techniques in real systems are described. Finally, spectrum sharing directions for 5G systems are analysed for different spectrum sharing techniques. It is concluded that licensed dedicated spectrum will continue to be the dominant spectrum usage method for 5G systems due to the possibility to control interference and guarantee coverage, while other spectrum sharing scenarios will act as complementary spectrum usage methods when beneficial.

Part II of this book presents new transmission and design techniques for 5G networks with a focus on physical-layer enabling techniques. It contains 11 chapters.

The chapter “Massive MIMO Communications” argues that every new network generation needs to make a leap in area data throughput, to manage the growing wireless data traffic. Massive MIMO technology can bring at least tenfold improvements in area throughput by increasing the spectral efficiency (bit/s/Hz/cell) while using the same bandwidth and density of base stations as in current networks. These extraordinary gains are achieved by equipping the base stations with arrays of a hundred antennas to enable spatial multiplexing of tens of user terminals.

This chapter explains the basic motivations and communication theory behind the massive MIMO technology and provides implementation-related design guidelines.

The chapter “Millimeter-Wave Mobile Communications” introduces key technologies of mmWave communications. Channel measurements show that mmWave signals suffer from much larger propagation losses and are suitable for small cell coverage. A hybrid network is presented where mmWave is used for capacity enhancement in hotspots, and a low-frequency network is applied for seamless coverage. Uniform air interface is a consequence to simplify the design between mmWave bands and low-frequency bands. Unified access and backhaul technique not only reduces the cost of backhaul but also can meet the requirement of 1000 times capacity enhancement over LTE systems.

The chapter “Non-Orthogonal Multiple Access (NOMA) for Cellular Future Radio Access” introduces state-of-the-art NOMA techniques and evaluates the low density spreading (LDS)-based system, which is a strong candidate for the next generation of mobile networks due to its well-known advantages compared to state-of-the-art techniques based on orthogonal frequency division multiple access (OFDMA). Furthermore, the effect of LDS parameters such as density factor and maximum number of users at each time instance on the sum rate is evaluated. The effect of irregularity on the complexity is also discussed. Moreover, it is shown that the loss of achievable rates which is caused by modulation can be compensated by using a suitable channel coding scheme.

The chapter “New Multicarrier Modulations for 5G” presents recent advances in filter bank multicarrier (FBMC) techniques and compares them with the conventional cyclic prefix (CP)-OFDM approach, in the context of 5G. After a brief description of some adaptations of CP-OFDM, FBMC combined with offset-QAM is considered, pointing out the crucial issue of subchannel equalisation. Then, an alternative approach is proposed, FBMC combined with pulse amplitude modulation (PAM). FBMC-PAM is an attractive option whenever asynchronous access and high level of out-of-band rejection are required. Finally, the case of nonoverlapping emitted symbols is considered, and a CP-less OFDM scheme with frequency domain equaliser in the receiver is included in the performance comparison.

The chapter “Fundamentals of Faster-than-Nyquist Signaling” presents the fundamentals of Faster-than-Nyquist (FTN) signalling. As originally introduced, FTN increases the bit rate in the signalling bandwidth by packing symbols closer in time, at the cost of introducing intersymbol interference (ISI). The chapter begins with the Euclidean distance properties of bandwidth-efficient pulses at FTN rates and describes receivers that mitigate the severe ISI. The FTN achievable information rate is compared with the Nyquist information rate for practical pulses. It then discusses the FTN extension to multicarrier systems with not only time packing but also subcarrier, optimising both the time and frequency packing.

The chapter “Generalized Frequency Division Multiplexing: A Flexible Multi-Carrier Waveform for 5G” aims to develop a unified air interface that can be configured on-the-fly to address emerging 5G applications. Apart from an ever-increasing demand for data rates, 5G is facing new applications such as Tactile

Internet and the Internet of Things. Being aligned with the whole concept of software-defined networking, this chapter introduces the multicarrier waveform termed generalised frequency division multiplexing (GFDM) as the basis for realising such a flexible physical design.

The chapter “Spectrally Efficient Frequency Division Multiplexing for 5G” focuses on novel multicarrier communication techniques, which share the common goal of increasing spectrum efficiency in future communication systems. In particular, a technology termed spectrally efficient frequency division multiplexing (SEFDM) is described in detail outlining its benefits, challenges, and trade-offs when compared to the current state of the art. A decade of research has been devoted to examining SEFDM from different angles: mathematical modelling, algorithm optimisation, hardware implementation, and system experimentation. The aim of this chapter is to therefore give a taste of this technology, and in doing so, the chapter concludes by outlining a number of experimental test beds which have been developed for the purpose of evaluating the performance of SEFDM in practical scenarios.

The chapter “Full-Duplex Wireless Communications for 5G” introduces full-duplex (FD) wireless communications for 5G, which enables simultaneous transmission and reception over the same frequency band. In this way, the spectral efficiency can be improved significantly compared with half-duplex (HD). However, there exists severe self-interference (SI), signal leakage from the local transmitter to its own receiver. Three different classes of SI mitigation techniques are presented in this chapter, i.e. propagation-domain SI suppression, analogue-domain SI cancellation, and digital-domain SI cancellation. Furthermore, the system performance of several FD schemes in several different application scenarios is presented. Theoretically, the spectral efficiency of FD bidirectional and cooperative communications can be doubled, while for cognitive radio networks, the FD-based protocol can achieve much better sensing performance than the traditional HD-based cognitive radio schemes.

The chapter “Device-to-Device Communications over 5G Systems: Standardization, Challenges and Open Issues” introduces one of the key enabling technologies at the heart of the 5G systems, namely, device-to-device (D2D) communications. The potential of D2D communication paradigm holding the promise to overcome the limitations of conventional cellular systems with very high bit rates, low delay, and low power consumption is illustrated. Starting from an overview of D2D communication technology, this chapter will browse through the main aspects that characterise the proximity services, with a view on the standardisation process, the challenges, and the open issues.

The chapter “M2M Communications in 5G” provides an overview of machine-type communications (MTC) within the context of 5G networks. The Internet of Everything foresees a hyperconnected World where humans, things, and machines will need to coexist together. They will be interconnected and Internet-connected via communication networks. In specific, the authors review the key novel challenges of MTC: what is new with regard to human-type traffic (HTC). They then analyse existing communication technologies and how suitable they are for MTC. Finally,

the authors identify key technology enablers being considered for the design of 5G networks and provide an outlook for the future.

The chapter “Design Techniques of 5G Mobile Devices in the Dark Silicon Era” is concerned with the design of the prospected 5G mobile communication system, which needs wide skills in wireless communication, analogue circuit design, embedded system, microwave technology, and so forth. System-level analyses, design space exploration, and performance trade-offs are some key steps that enable the design of low-cost, energy-efficient, ubiquitous, and flexible transceiver. This chapter provides comprehensive design techniques for 5G mobile communication in the dark silicon era using More than Moore technology (MtM).

Part III of this book focuses primarily on the networking and application layer techniques for 5G networks, which includes 12 chapters.

The chapter “Ultra-Dense Network Architecture and Technologies for 5G” presents the ultra-density network (UDN), which is the most promising way to meet the ultrahigh area capacity requirement for 5G. The content of this chapter includes characters of UDN scenarios, network architecture design, and key technologies like flexible networking, wireless backhauling, multi-RAT coordination, mobility management, interference management, and radio resource management.

The chapter “5G RAN Architecture: C-RAN with NGFI” describes cloud radio access networks (C-RAN), which are viewed as one of the key RAN architectures for 5G networks, with evolved architecture based on a newly designed fronthaul interface, dubbed the next-generation fronthaul interface (NGFI). The design principles and the challenges of NGFI are introduced. A prototype is further developed to verify the applicability of NGFI-based C-RAN.

The chapter “User-Centric Wireless Network for 5G” addresses the concept of user-centric wireless network for 5G from the perspective of fulfilling multiple user experience requirements in 5G. Four key technical directions are studied based on a gap analysis between LTE technology and 5G requirements, i.e. user-centric 5G access network architecture design, flexible functionality and deployment, smart user and traffic awareness and management, and high-efficient low-cost network operation. These key technologies work together with cross-layer and end-to-end solutions to provide the user-centric 5G ecosystem.

The chapter “Energy Harvesting Based Green Heterogeneous Wireless Access for 5G” is concerned with the issues of energy harvesting for future 5G cellular systems. A feasible and efficient method to tackle this issue is to let the communication systems harvest energy from renewable energy sources instead of fossil fuels. However, by employing the energy harvesting (EH) technique, the instability of renewable energy resources introduces new challenges on the design of the upcoming 5G systems. This chapter focuses on uplink access schemes and power allocations for EH-based heterogeneous networks. First, a heterogeneous access model incorporating EH-based mobile users is proposed and followed by a throughput maximisation framework. Then, by classifying transmission policies into two main categories (i.e. single-channel vs. multichannel scenarios), the proposed framework is concretised under various practical conditions, including

the availability of central control, causality of harvested energy, channel state information, and others.

The chapter “Resource Management in Sustainable Green HetNets with Renewable Energy Sources” investigates the energy sustainability performance of a green HetNet where the small cell base stations (SBSs) are powered by green energy. Specifically, we first develop an analytical framework to study the energy sustainability of each SBS. Then, we propose a distributed admission control strategy at SBSs striking a balance between resource utilisation and energy sustainability. Extensive simulations validate the analytical framework and demonstrate that relaxing the admission control criteria can improve resource utilisation when the energy is abundant, but may significantly degrade resource utilisation instead when the energy comes short due to poor sustainability performance.

The chapter “Resource Allocation for Cooperative D2D Communication Networks” studies various resource allocation policies for cooperative device-to-device (D2D) communications in systems operating under OFDMA or cognitive radio architectures. A variety of system models are explored, wherein additional features, such as, packet storage, energy harvesting, and cognitive radio capabilities, are incorporated at the user devices so as to enable cooperative D2D communications. Computationally efficient solutions are provided for multiple resource optimisation problems including power allocation, subcarrier allocation, subcarrier pairing, and relay selection. Simulation results demonstrate that the sum-throughput performance can be improved whenever the user devices are equipped with cooperative D2D capabilities.

The chapter “Fog Computing and Its Applications in 5G” explains the emergence of fog computing as a promising, practical, and efficient solution tailored to serving mobile traffics. Fog computing deploys highly virtualised computing and communication facilities at the proximity of mobile users. Dedicated to serving the mobile users, fog computing explores the predictable service demand patterns of mobile users and typically provides desirable localised services accordingly. It can provide mobile users with the demanded services via low-latency and short-distance local connections. The authors introduce the main features of fog computing and describe its concept, architecture, and design goals. Lastly, they discuss the potential research issues from the perspective of 5G networking.

The chapter “A Conceptual 5G Vehicular Networking Architecture” shows how 5G communication systems will help to enable connected future cars to implement automated functions in short term and fully autonomous operation in long term. The authors review the well-known existing communication technologies for connected cars and analyse their shortcomings. Towards this end, they outline the innovation areas that 5G aims to address in order to mitigate the limitations of the current technologies.

The chapter “Communications Protocol Design for 5G Vehicular Networks” provides an overview on existing standards in vehicular networking and highlights new emerging trends towards an integrated infrastructure based on the interworking of heterogeneous technologies. Next-generation mobile vehicular networks are first characterised by providing an insight on relevant stable standards in wireless

communication technologies, with a special focus on heterogeneous vehicular networks. Furthermore, the chapter discusses a general framework supporting opportunistic networking scheme and outlines novel application and use cases based on social- and context-awareness paradigms.

The chapter “Next-Generation High-Efficiency WLAN” centres around the topic of next-generation high-efficiency WLAN technology. With the increasing demands for WLAN and the deployment of carrier-WiFi networks, the number of WiFi public hotspots worldwide is expected to increase dramatically. To face this huge increase in the number of densely deployed WiFi networks, and the massive amount of data to be supported by these networks in indoor and outdoor environments, it is necessary to improve the current WiFi standard and define specifications for high-efficiency wireless local area networks (HEWs). This chapter introduces emerging HEW technology, including its typical use cases, environments, and potential techniques that can be applied for HEWs. The typical HEW use cases are first given, followed by an analysis of the main requirements from these use cases and environments. Then, potential techniques, including enhanced medium access and spatial frequency reuse, are presented and discussed.

The chapter “Shaping 5G for the Tactile Internet” investigates the topic of the Tactile Internet, which is expected to have a massive impact on business and society. It has the potential to revolutionise almost every segment of the society by enabling wireless control and remote operation in a range of scenarios. The next-generation (5G) mobile communication networks will play an important role in realising the Tactile Internet. This chapter investigates the interesting area of 5G and Tactile Internet intersection. Key requirements for the Tactile Internet, from a networking perspective, have been identified, after introducing exciting Tactile Internet applications. The chapter covers several technical issues and challenges in shaping 5G networks for realising the vision of the Tactile Internet. The most important challenge would be to ensure tight and scalable integration of various technological solutions into a single network.

We would like to thank all the authors who submitted their research work to this book. We would also like to acknowledge the contribution of many experts who have participated in the review process and offered comments and suggestions to the authors to improve their works. We would also like to express our sincere appreciation to the editors from Springer for their support and assistance during the development of this book.

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Part I

Overview of 5G Networks

An Overview of 5G Requirements

Dajie Jiang and Guangyi Liu

Abstract Mobile Internet and IoT (Internet of Things) are the two main market drivers for 5G. There will be a massive number of use cases for Mobile Internet and IoT, such as augmented reality, virtual reality, remote computing, eHealth services, automotive driving and so on. All these use cases can be grouped into three usage scenarios, i.e., eMBB (Enhanced mobile broadband), mMTC (Massive machine type communications) and URLLC (Ultra-reliable and low latency communications). Eight key capabilities including peak data rate, latency and connection density, etc., are defined to meet the requirements of usage scenarios. Based on the usage scenarios, several typical deployment scenarios including indoor hotspots, dense urban, urban macro, rural and high-speed scenarios are specified, together with the detailed technical requirements for 5G. Both the deployment scenarios and technical requirements are essential guidance for 5G technical design.

1 Introduction

A mobile and connected society is emerging in the near future, which is characterized by a tremendous amount of growth in connectivity, traffic volume and a much broader range of use scenarios [1]. Some typical trends are summarized as follows:

- Explosive growth of data traffic: There will be an explosive growth in traffic. The global data traffic will increase by more than 200 times from 2010 to 2020, and about 20,000 times from 2010 to 2030;
- Great increase in connected devices: While smart phones are expected to remain as the main personal devices, the number of other kinds of devices, including wearable devices and MTC devices will continue to increase;
- Continuous emergence of new services: Different kinds of services, e.g., services from enterprises, from vertical industries and Internet companies, etc. will be exploited.

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The fifth-generation (5G) mobile communications system will emerge to meet new and unprecedented demands beyond the capability of previous generations of systems.

There are two phases of 5G requirements research by different organizations. Phase 1 focuses on 5G use cases and high-level key capabilities of 5G networks, and can be regarded as the 5G vision stage. In Phase 1, ITU has released the vision recommendation [2] and defined the key capabilities of 5G. 3GPP started the smarter program [4] and studied 5G use cases and requirements. NGMN completed a 5G whitepaper and defined a large number of 5G use cases and requirements [5]. IMT-2020 (5G) Promotion Group released the 5G vision and requirements whitepaper in May 2014 [1], which aims to contribute to the ITU-R work in Phase 1. Phase 2 focuses on 5G deployment scenarios and detailed technical requirements. There are two important reports in Phase 2. One is the IMT-2020 technical performance requirements from ITU-R which will be completed by February 2017 [3], while the other is the scenarios and requirements technical report from 3GPP which will be completed in March 2017 [6]. NGMN has started the relevant work at the beginning of 2015 and drafted several liaisons to 3GPP and ITU by March 2016 [7, 8]. IMT-2020 (5G) PG plans to complete the evaluation scenarios and the KPI report in the first half of 2016, and will have an impact on the work of ITU and 3GPP in Phase 2 (Fig. 1).

The rest of this chapter is organized as follows. The outcomes of Phase 1, i.e., 5G use cases and high-level key capabilities are introduced in Sects. 2 and 3, respectively. The latest status of Phase 2 including deployment scenarios

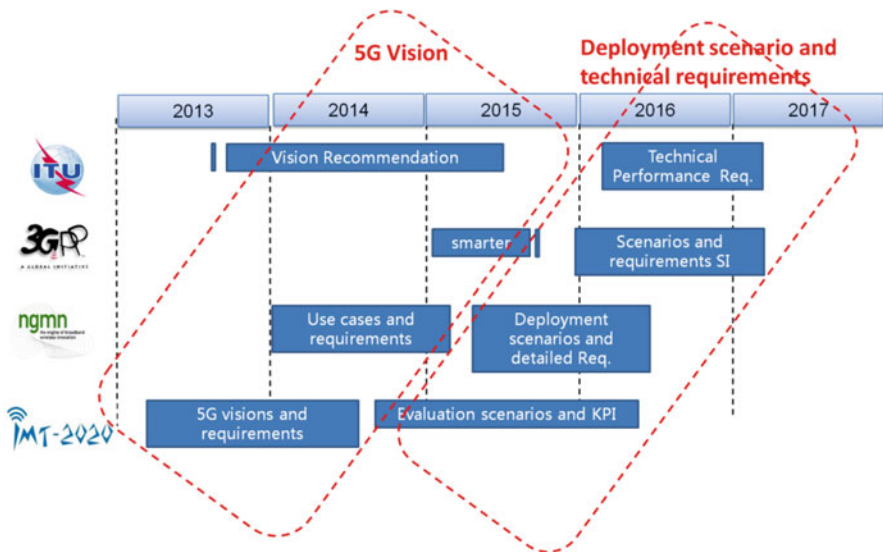


Fig. 1 Overview of 5G requirements research by different organizations

and detailed technical requirements are presented in Sects. 4 and 5, respectively. Section 6 presents the operational requirements, while Sect. 7 draws concluding remarks.

2 Use Cases and Challenges

It is foreseen that there will be a huge number of use cases in the upcoming 5G era. In [1], an overall vision for 5G life is illustrated in Fig. 2. 5G will penetrate into every single element of our future society and create an all-dimensional, user-centered information ecosystem. 5G will break the limitation of time and space to enable an immersive and interactive user experience. 5G will also shorten the distance between human and things, and implement a seamless integration to achieve easy and smart interconnection between people and all things. 5G will enable us to realize the vision—“Information is a finger away, and everything will be kept in touch”.

There are a great many use cases proposed by different organizations [1, 9–11]. Mobile Internet and the Internet of Things (IoT) are the two main market drivers in the future development of mobile communications [1], and they will trigger a large range of use cases.



Fig. 2 Overall vision of 5G

2.1 Use Cases of Mobile Internet

Mobile Internet is disrupting the traditional business model of mobile communications, enabling unprecedented user experiences and making a profound impact on every aspect of people's work and life. Looking ahead to 2020 and beyond, mobile Internet will promote the continued evolution of the way human interacting information, and provide users with ultimate experience through more immersive services including but not limited to:

- Video services, such as immersive Ultra High Definition (UHD) and three-dimensional (3D) video
- Augmented reality
- Virtual reality
- Video/photo sharing in stadium/open air gathering
- Online gaming applications
- Mobile cloud/desktop cloud
- Tactile Internet
- Remote computing
- 3D connectivity: aircraft
- 3D connectivity: drones
- Collaborative robots
- Broadcast-like services, like local, regional and national news and information
- Smart office

The future development of Mobile Internet will trigger the growth of mobile traffic by a magnitude of thousands in the future, and promote a new wave of upgrades and a revolution in mobile communications technologies and the telecommunications industry as a whole. Looking ahead to 2020 and beyond, there will be an explosive growth in mobile data traffic. It is estimated by IMT-2020 (5G) Promotion Group that the global mobile data traffic will grow by more than 200 times from 2010 to 2020, and by nearly 20,000 times from 2010 to 2030. In China, the growth rate is projected to be even higher, with mobile data traffic expected to grow by more than 300 times from 2010 to 2020 and by more than 40,000 times from 2010 to 2030. For developed cities and hotspots in China, the growth of mobile data traffic will exceed the average projected. For example, from 2010 to 2020 in Shanghai, the mobile data traffic is projected to grow by 600 times. In Beijing and during this same period, it is estimated that hotspot traffic may grow by up to 1000 times. The above estimation is shown in Fig. 3.

There is some traffic anticipation work by ITU-R and the results are detailed in Report ITU-R M.[IMT.2020BEYOND TRAFFIC] [12]. This report contains global IMT traffic estimates beyond 2020 from several sources. These estimates anticipate that global IMT traffic will grow by 10–100 times from 2020 to 2030. The main drivers behind the anticipated traffic growth include increased video usage, device proliferation and application uptake. These are expected to evolve over time, and this evolution will differ between countries due to social and economic differences.

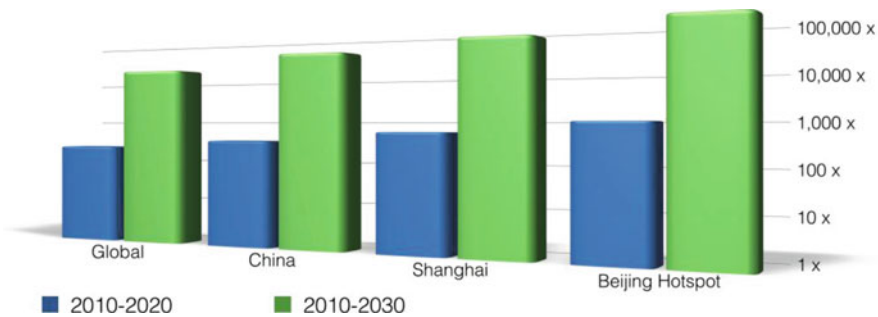


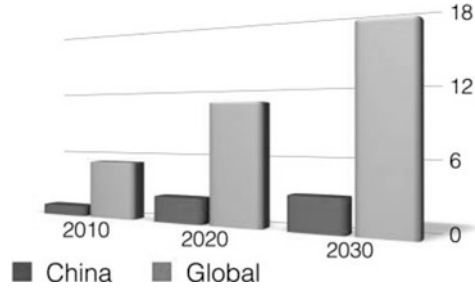
Fig. 3 2010–2030 growth of mobile data traffic

Traffic asymmetry aspects for this period are also presented by ITU-R. It is observed that the current average traffic asymmetry ratio of mobile broadband is in favor of the downlink, and this is expected to increase due to a growing demand for audio-visual content.

Mobile Internet is aiming at people-oriented communications with a focus on the user experience. Towards 2020 and beyond, the increasing popularity of ultra-high definition (UHD), 3D and video immersion will significantly drive up the data rates. For example, with a hundredfold compression, the transmission of 8K (3D) video will require a transmission rate close to 1 Gbps. Services, such as augmented reality, desktop cloud, and online gaming will not only pose a challenge to uplink and downlink data transmission rates but also generate a stringent requirement for the so-called “imperceptible latency”. In the future, vast amounts of individuals and office data will be stored in the cloud. Such massive data activity will require transmission rates to be comparable to optical fiber communications, which will lead to enormous traffic challenges for mobile communications networks particularly in hotspot areas. Over-the-top (OTT) services, such as social networking, will be counted among leading applications going forward, and the associated frequently-occurring small packets will devour signaling resources. At the same time, consumers will continue to demand better experiences on mobile communications wherever they are. A consistent service experience is expected in all scenarios, including ultra-dense scenarios such as stadiums, open-air gatherings and concerts, and high-speed moving scenarios such as high-speed trains, vehicles and subways.

The total number of devices connected by global mobile communications networks will reach 100 billion in the future. By 2020, it is predicted that the number of mobile terminals around the world will surpass ten billion, of which China will contribute over two billion, as shown in Fig. 4.

Fig. 4 2010–2030 growth of mobile device



2.2 Use Cases of Internet of Things

The IoT has extended the scope of mobile communications services from interpersonal communications to interconnection between things (smart devices), and between people and things, allowing mobile communications technologies to penetrate into broader industries and fields. By 2020 and beyond, applications such as mobile health, Internet of Vehicles (IoV), smart home, industrial control, and environmental monitoring will drive the explosive growth of IoT applications, facilitating hundreds of billions of devices to connect to a network creating a true “Internet of Everything”. This will give rise to emerging industries of a unprecedented scale and instill infinite vitality to mobile communications. Meanwhile, the massive number of interconnected devices and the diversified IoT services will also pose new challenges to mobile communications. The potential IoT use cases include:

- Smart Grid and critical infrastructure monitoring
- Environmental monitoring
- Smart agriculture
- Smart metering
- eHealth services
- Remote object manipulation like remote surgery
- Automotive driving/Internet of vehicles
- Smart wearables, like sports and fitness
- Sensor networks
- Mobile video surveillance
- Smart cities
- Smart transportation
- Smart home
- Industrial control

IMT-2020 (5G) Promotion Group has estimated the number of IoT devices in future years as shown in Fig. 5. It is projected that the total number of devices connected by the global mobile communications network will reach 100 billion in the future. By 2020, the number of mobile terminals around the world will surpass ten billion, of which China will contribute over two billion. The number of IoT

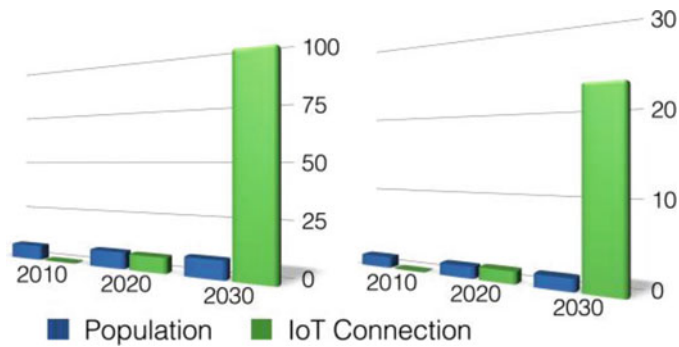


Fig. 5 2010–2030 growth of IoT connections

connections will also expand rapidly, reaching the size of global population of seven billion by 2020. By 2030, the number of global IoT connections will reach 100 billion. Among all types of terminals, smart phones will contribute most traffic, and IoT terminals will contribute less, even though the number of devices is much larger.

IoT is focused on communications between things and between things and people, involving not only individual users, but also a large number of various vertical industrial customers. IoT services types and relevant requirements of IoT services are very diverse. For services such as the smart home, smart grid, environmental monitoring, smart agriculture, and smart metering, the network will be required to support a massive amount of device connections and frequently-occurring small data packets. Services like video surveillance and mobile health will have strict requirements on transmission rates, while services such as IOV and industrial control will demand millisecond-level latency and nearly 100 % reliability. In addition, many IoT devices may be deployed in remote, or in areas where transmission losses can be a problem, such as indoor corners, basements and tunnels. Therefore, the coverage of mobile communications networks need to be further enhanced. In order to penetrate into more IoT services, 5G should be more flexible and more scalable, to support massive device connections and meet diverse user requirements.

Users expect better and yet more cost-effective services and experiences with mobile Internet and IoT. In addition to satisfying cost and experience demands, 5G will also need to meet extremely high security requirements, particularly for services such as e-banking, security monitoring, safe driving, and mobile health. 5G will also be able to support lower power consumption to build greener mobile communication networks and to enable much longer terminal battery life, especially for some IoT devices.

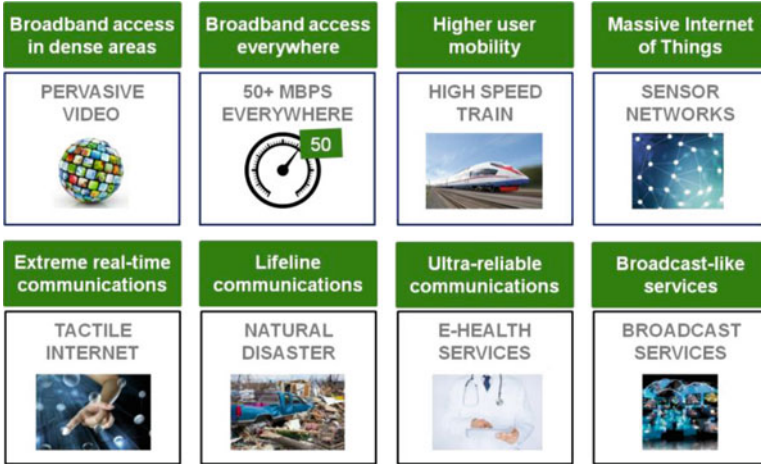


Fig. 6 5G use case families and related use case examples

2.3 Classification of 5G Use Cases

5G will support a large variety of use cases which are emerging now or will emerge in the future. Different use cases have varying characteristic and requirements. It is helpful to group countless emerging use cases into several use case families. Use cases in each use case family share similar characteristic and requirements.

NGMN has developed 25 use cases for 5G as representative examples, which are grouped into eight use case families. The following diagram [5] illustrates the eight use case families with one example use case given for each family, and these families and their corresponding use case examples are described in Fig. 6.

ITU-R has concluded three usage scenarios (use case groups) addressing different use case characteristics in Fig. 7 [2]:

- Enhanced mobile broadband: Mobile broadband addresses human-centric use cases for access to multi-media content, services and data. The demand for mobile broadband will continue to increase, leading to enhanced mobile broadband. The enhanced mobile broadband usage scenario will come with new application areas and requirements in addition to existing mobile broadband applications for improved performance and an increasingly seamless user experience.
- Ultra-reliable and low latency communications: This use case has stringent requirements for capabilities such as throughput, latency and availability. Some examples include wireless control of industrial manufacturing or production processes, remote medical surgery, distribution automation in a smart grid, transportation safety, etc.

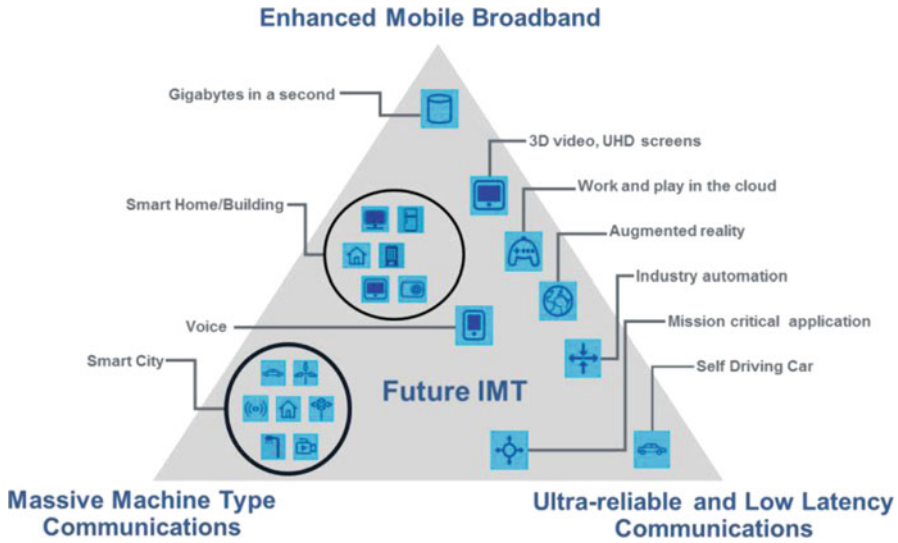


Fig. 7 5G usage scenarios

- **Massive machine type communications:** This use case is characterized by a very large number of connected devices typically transmitting a relatively low volume of non-delay-sensitive data. Devices are required to be of low cost, and have a very long battery life.

Similarly, IMT-2020 (5G) Promotion Group has proposed four technical scenarios for 5G in Fig. 8 which are well in line with the three usage scenarios from ITU-R. The main difference is that the eMBB scenario from ITU-R is divided into two technical scenarios, i.e., the seamless wide-area coverage scenario and high-capacity hot-spot scenario.

For the seamless wide-area coverage scenario, seamless coverage and medium to high mobility are desired, with much improved user data rate compared to existing data rates. However, the data rate requirement may be relaxed compared to the hotspot scenario. For the high-capacity hot-spot scenario, i.e., for an area with high user density, very high traffic capacity is needed, while the requirement for mobility is low and the user data rate is higher than that of wide area coverage. For the wide area coverage case, seamless coverage and medium to high mobility are desired, with much improved user data rate compared to existing data rates. However, the data rate requirement may be relaxed compared with the hotspot.

Figure 9 shows the mapping from the eight use case families proposed by NGMN to the three usage scenarios defined by ITU-R. The eMBB usage scenario consists of broadband access in dense areas, broadband access everywhere, higher user mobility and broadcast-like services. URLLC consists of extreme real-time communications, lifeline communications and ultra-reliable communications. mMTC corresponds to massive Internet of Things.



Fig. 8 Technical scenarios for 5G

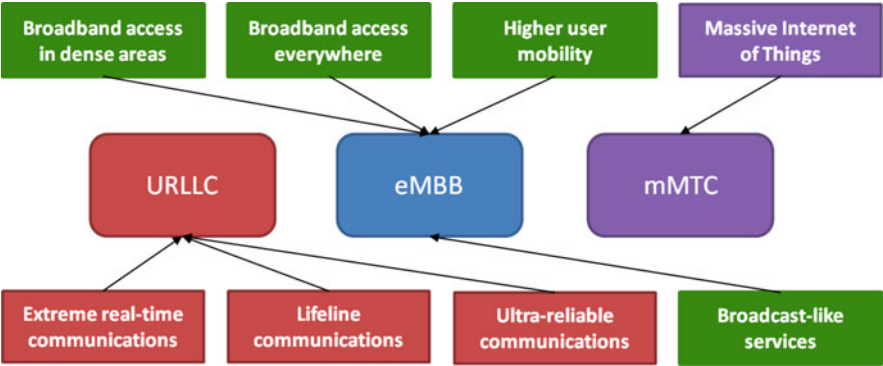


Fig. 9 Mapping eight use case families to three use scenarios

3 High-Level Key Capabilities

IMT-2020 (5G) PG groups the 5G high-level requirements into several performance indicators and efficiency indicators. The key performance indicators for 5G include the user experienced data rate, connection density, end-to-end delay, traffic volume density, mobility, and peak data rate. Their definitions are listed in Table 1. The value for each performance indicator and the relevant scenario are illustrated in Fig. 10.

Table 1 5G Performance indicators

Performance indicators	Definition
User experienced data rate (bps)	Minimum achievable data rate for a user in real network environment
Connection density (/km ²)	Total number of connected devices per unit area
End-to-end latency (ms)	Duration between the transmission of a data packet from the source node and the successful reception at destination node
Traffic volume density (bps/km ²)	Total data rate of all users per unit area
Mobility (km/h)	Relative speed between receiver and transmitter under certain performance requirement
Peak data rate (bps)	Maximum achievable data rate per user

Several problems are anticipated if today’s networks are used to handle the explosive development of mobile Internet and IoT:

- The energy efficiency level, overall cost per bit and complexity of network deployment and maintenance cannot effectively handle 1000 times traffic growth and the massive number of connected devices in the next decade;
- Co-existence of multiple radio access technologies (RAT) causes increased complexity and degraded user experience;
- Existing networks can not realize accurate monitoring of network resources and effective awareness of services, and therefore they cannot intelligently fulfill the diversified requirements of future users and services;
- Widely distributed and fragmented spectrum will cause interference and co-existence complexity.

To solve these problems, 5G should have the following capabilities to achieve sustainability. In terms of network construction and deployment, 5G networks need to:

- Provide higher network capacity and better coverage, while decreasing the complexity and cost of network deployment, especially the deployment of ultra-dense networks.
- Have a flexible and scalable architecture to adapt to the diverse needs of users and services.
- Make flexible and efficient use of various spectrum resources, including paired and unpaired spectrum, re-farmed spectrum and new spectrum, low-frequency and high-frequency bands, and licensed and unlicensed bands.
- Have stronger device-connection capabilities to deal with the access requirements of huge amounts of IoT devices.

In terms of operation and maintenance (O&M), 5G needs to:

- Improve network energy efficiency and the O&M cost-per-bit to cope with data traffic growth and the diverse needs of various services and applications.

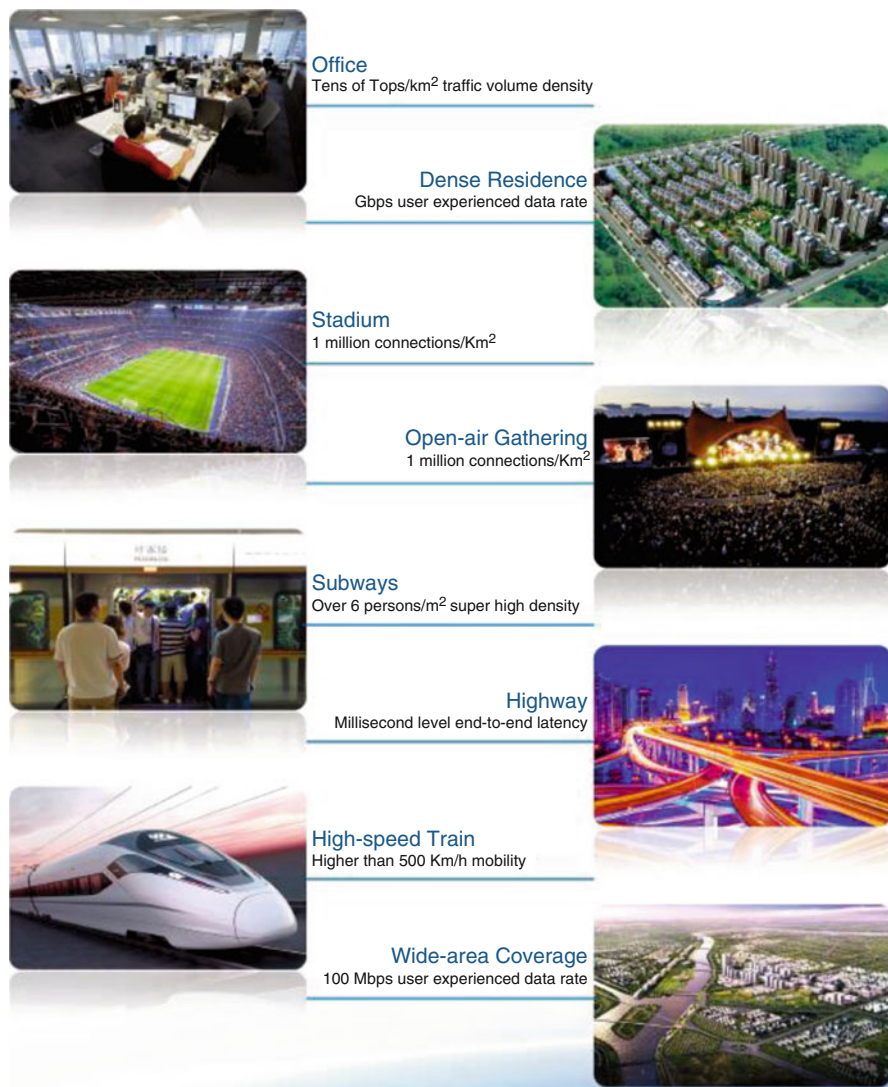


Fig. 10 Challenging scenarios and performance indicators

- Reduce the complexity caused by the co-existence of multiple radio access technologies, network upgrades, and the introduction of new features and functions, to improve users' experience.
- Make intelligent optimization based on awareness of users behaviors and services contents
- Provide a variety of network security solutions to meet the needs of all types of devices and services of mobile internet and IoT.

Table 2 5G Key efficiency indicators

Efficiency indicators	Definition
Spectrum efficiency (bps/Hz/cell or bps/Hz/km ²)	Data throughput per unit of spectrum resource per cell (or per unit area)
Energy efficiency (bit/J)	Number of bits that can be transmitted per joule of energy
Cost efficiency (bit/¥)	Number of bits that can be transmitted per unit cost

Spectrum utilization, energy consumption and cost are the three key factors which must be addressed in sustainable mobile communication networks. In order to achieve sustainability, 5G needs to make significant improvements in the following aspects (Table 2):

- Spectrum efficiency: 3–5 times
- Energy efficiency: 100+ times
- Cost efficiency: 100+ times

5G systems must dramatically outperform previous generation systems. 5G should support

- User experienced data rate: 0.1–1 Gbps
- Connection density: one million connections per square kilometer
- End-to-end latency: millisecond level
- Traffic volume density: tens of Gbps per square kilometer
- Mobility: higher than 500 km per hour
- Peak data rate: tens of Gbps

Among these requirements, the user experienced data rate, connection density and end-to-end latency are the three most fundamental ones. Meanwhile, 5G needs to significantly improve the efficiency of network deployment and operations. Compared with 4G, 5G should have 3–5 times improvement on spectrum efficiency and more than 100 times improvement on energy and cost efficiency.

The performance requirements and efficiency requirements define the key capabilities of 5G, which can be illustrated as a “blooming flower” depicted in Fig. 11. The petals and leaves rely on each other. The petals represent the six key capabilities in terms of performance and can fulfill the diverse requirements of future services and scenarios. The leaves represent the three key capabilities in terms of efficiency, and can guarantee the sustainable development of 5G. The top of each petal means the maximum value of the corresponding capability.

The key capabilities of IMT-2020 defined by ITU-R are shown in Fig. 12, compared with those of IMT-Advanced. The values for each key capability are shown in Table 3.

All key capabilities may to some extent be important for most use cases, and the relevance of certain key capabilities may be significantly different, depending on the use cases/scenario. The importance of each key capability for the usage scenarios